

Another important phase in the development of AI could be traced back to the first theoretical analysis of AI by Allan Turing. The Turing test proposed by him suggests one of the earliest methods to test artificial intelligence. The test may not be relevant today but it was one of the first attempts to define artificial intelligence. It can be summarised as follows. Behind a curtain, there is either a person named A or a machine named B. You can talk with him/it for say 30 minutes, after which you should be able to identify whether you were conversing with a person or a machine. With the powerful chatbots of today, it is easy to see that such a test will not identify true AI. But in the early 1950s this did give a theoretical framework to understand AI.

In the late 1950s, a programming language called Lisp was developed by John McCarthy. This was one of the earliest high-level programming languages. Until then, machine languages and assembly languages (notoriously difficult to use) were used for computer programming. A very powerful machine, a very powerful language to communicate with it, and computer scientists being the optimists and dreamers made the hope for artificially intelligent machines reasonable for the first time in the history of humanity. In the early 1960s, the hopes for artificially intelligent machines were at their peak. Of course, there were a lot of developments in the computing world, but were there any wonders? Sadly, no. So, the 1960s saw hope and despair about AI for the first time. However, computer science was developing at a pace that is unparalleled in any field of human endeavour.

Then came the 70s and the 80s, where algorithms played a major role. A lot of new and efficient algorithms were developed during this period. The famous textbook 'The Art of Computer Programming' written by Donald Knuth (I request you to read about this person, in the world

of computer science he is either Gauss or Euler) had its first volume published in the late 1960s marking the beginning of the algorithmic era. A lot of general and graph algorithms were developed during these years. Further, programming based on artificial neural networks was also introduced during this time. Though McCulloch and Pitts were the pioneers in proposing artificial neural networks as far back as the 1940s, it took decades to become a mainstream technique. Today, deep learning is almost exclusively based on artificial neural networks. Such developments in the field of algorithms led to a resurgence in research in the field of artificial intelligence in the 1980s. However, this time, limitations in communication and computing powers derailed progress in this area. For a second time, AI couldn't live up to the huge expectations it had generated. Then came the 90s, Y2K, and the present day. Once again there is a lot of enthusiasm and hope about the positive impact of AI based applications.

Based on our discussion, it is easy to see that there were at least two other occasions in the digital era when AI was promising. On both those occasions AI didn't live up to its expectations. Is the current enthusiasm around AI a similar one? Of course, this is a very difficult question to answer. But in my personal opinion, this time AI is going to make a huge impact. But what are the factors that made me make such a prediction? First, very powerful computing devices are cheaply available nowadays. Unlike in the 1960s or 1980s, when a handful of such powerful machines were available, we now have them in the millions and billions. Second, there are large data sets available today for training AI and machine learning based applications. Imagine an AI engineer working in the field of digital image processing in the 90s. How many digital images were available for training his/her algorithms? Well, maybe a

few thousands or tens of thousands. A cursory search on the Internet will tell you that the data science platform Kaggle (a subsidiary of Google) itself has more than ten thousand data sets. The large amount of data produced on the Internet every day makes training of algorithms a lot easier. Third, high speed connectivity has made it easy to collaborate with large groups. Imagine the difficulty of computer scientists collaborating with each other until the first decade of the 21st century. The present day speed of the Internet has made collaborative AI based projects like Google Colab, Kaggle, Project Jupyter, etc, a reality. Owing to these three factors, I believe this time AI is here for good and will have some excellent applications.

More about matrices

Now that we are somewhat familiar with the history of AI, it is time to revisit one of the main themes of this tutorial on AI — matrices and vectors. In the last article, we began discussing these. This time we are going to dive a little deeper into the world of matrix theory. I will begin by pointing your attention to Figures 1 and 2, which show eight matrices (A, B, C and D, and matrices E, F, G and H). Why so many matrices in a tutorial on AI and machine learning? First, as mentioned in the previous article in this series, matrices are the core of linear algebra, and linear algebra is the heart of machine learning, if not

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 8 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 11 & 22 & 33 \end{bmatrix}$$

Figure 1: Matrices A, B, C and D

$$E = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$F = \begin{bmatrix} 11 & 22 & 33 \\ 77 & 88 & 99 \end{bmatrix}$$

$$G = \begin{bmatrix} 11 \\ 22 \\ 33 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$$

Figure 2: Matrices E, F, G and H